Jingchu Gai

gaijingchu@gmail.com · (+1) 412-992-9467

Education

Carnegie Mellon University, Machine Learning Department, Pittsburgh, PA, USA — 2025 – Present

Ph.D. in Machine Learning
Advisors: Prof. Andrej Risteski, Prof. Aditi Raghunathan

Peking University, School of Mathematical Sciences, Beijing, China — 2021 – 2025

Bachelor of Science in Mathematics
Advisor: Prof. Liwei Wang
Overall GPA: 3.957/4.00
Ranking: 1/217

Research Interests

- LLM reasoning and post-training: reinforcement learning for LLMs, entropy control, mitigating sharpening and premature confidence
- Theoretical foundations of reinforcement learning and imitation learning, robust multi-agent RL
- Graph representation learning: GNN expressiveness, graph theory, algebraic graph theory

Publications

** denotes equal contribution; † denotes corresponding author*

- **Beyond Weisfeiler-Lehman: A Quantitative Framework for GNN Expressiveness** — **ICLR 2024**
  Bohang Zhang, **Jingchu Gai***, Yiheng Du, Qiwei Ye, Di He, Liwei Wang.*
  `https://arxiv.org/abs/2401.08514`
  ICLR 2024 Best Paper Honorable Mention, Oral
  International Conference on Learning Representations, 2024
- **Homomorphism Expressivity of Spectral Invariant Graph Neural Networks** — **ICLR 2025**
  ***Jingchu Gai**, Yiheng Du, Bohang Zhang, Haggai Maron, Liwei Wang.*
  `https://arxiv.org/abs/2503.00485`
  ICLR 2025 Oral
  International Conference on Learning Representations, 2025
- **Demystifying Entropy Control in LLM RL Training: Theoretical Analysis and Dynamic Scheduling** — **ICML 2026**
  Jingchu Gai, Guanning Zeng*, Huaqing Zhang, Han Zhong, Yige Hong, Andrej Risteski, Aditi Raghunathan.*
  ICML 2026 Spotlight
  International Conference on Machine Learning, 2026
- **Differential Smoothing Mitigates Sharpening and Improves LLM Reasoning** — **ICML 2026**
  Jingchu Gai, Guanning Zeng*, Huaqing Zhang*, Aditi Raghunathan.*
  `https://arxiv.org/abs/2511.19942`
  International Conference on Machine Learning, 2026
- **Towards Solving the Gilbert–Pollak Conjecture via Large Language Models** — **ICML 2026**
  *Yisi Ke, Tianyu Huang, Yankai Shu, Di He, **Jingchu Gai†**, Liwei Wang†.*
  `https://arxiv.org/abs/2601.22365`
  International Conference on Machine Learning, 2026
- **Breaking the Curse of Multiagency in Robust Multi-Agent Reinforcement Learning** — **ICML 2025**
  Laixi Shi, **Jingchu Gai***, Eric Mazumdar, Yuejie Chi, Adam Wierman.*
  `https://arxiv.org/abs/2409.20067`
  International Conference on Machine Learning, 2025
- **Understanding and Mitigating Premature Confidence for Better LLM Reasoning** — **COLM 2026**
  Jingchu Gai, Guanning Zeng*, Christina Baek*, Chen Wu*, J. Zico Kolter, Andrej Risteski, Aditi Raghunathan.*

https://arxiv.org/abs/2605.24396
Conference on Language Modeling, 2026

- **Towards Understanding the Benefits of Online Imitation Learning** **ICLR 2026 Workshop**
  Huaqing Zhang, **Jingchu Gai***, Jiwoo Kim, Bingbin Liu, Andrej Risteski.*
  The 1st Workshop on Scaling Post-training for LLMs
- **Momentum Streams for Optimizer-Inspired Transformers** **Preprint**
  Jingchu Gai*, *Nai-Chieh Huang*, Jiayun Wu* (authors in alphabetical order).*
  https://arxiv.org/abs/2605.24425
- **Lossless Anti-Distillation Sampling** **Preprint**
  *Zhihang Diao, **Jingchu Gai**, Xinyu Ai, Zihao Zhang, Zhonghao He, Di He.*
  https://arxiv.org/abs/2605.18829
- **Taming the Curses of Multiagency in Robust Markov Games with Large State Space through Linear Function Approximation** **Preprint**
  Jingchu Gai, *Laixi Shi.*
  https://arxiv.org/abs/2605.03125

Honors and Awards

ICLR 2024 Best Paper Honorable Mention (As first author)	2024
SenseTime Scholarship (Awarded to 25 undergraduates across China talented in AI)	2024
National Scholarship of China (The highest honor awarded to undergraduates in China)	2024
National Scholarship of China (The highest honor awarded to undergraduates in China)	2023
Merit student of Peking University	2024
Merit student of Peking University	2023
First Prize in the 14th Chinese College Students' Mathematical Contest	2023
Winner Prize in the 14th Yau Contest	2023
Academic Merit student of Peking University	2022
Scholarship of freshman of Peking University	2021
First Prize in the 37th Chinese Physics Olympiad	2020

Selected Coursework

- **Analysis:** Mathematical Analysis I (94), Mathematical Analysis II (98), Mathematical Analysis III (98) Theory of Functions of Complex Variables (99) Ordinary Differential Equation (96)
- **Algebra:** Advanced Algebra I (93), Advanced Algebra II (99), Abstract Algebra (97)
- **Programming:** Introduction to Computation (98.5), Data Structure and Algorithms (99)
- **Geometry:** Geometry (96.5)
- **Statistics:** Probability Theory (98), Mathematical Statistics (100), Statistical Thinking (99)
- **Others:** Game Theory (100)

Graduate Courses

- Applied Stochastic Process (99)
- High-Dimensional Probability (95)
- Statistical Models and Computing Methods (97.3)
- Mathematical Methods of Classical Mechanics (95)

Skills

- Programming Languages: Python, C++
- TOEFL iBT: 109/120 (Reading 30, Listening 30, Speaking 23, Writing 26)
- GRE General: 335/340 (Verbal 166, Quantitative 169)